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Portfolio Risk and Return: Part I

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PORTFOLIO OF MANY RISKY ASSETS

Learning Outcomes

- calculate and interpret portfolio standard deviation
- describe the effect on a portfolio's risk of investing in assets that are less than perfectly correlated

In the previous section, we discussed how the correlation between two assets can affect the risk of a portfolio and the smaller the correlation the lower is the risk. The above analysis can be extended to a portfolio with many risky assets (N). Recall the previous equations for portfolio return and variance:

$$E(R_p) = \sum_{i=1}^{N} w_i E(R_i), \quad \sigma_P^2 = \left(\sum_{i=1}^{N} w_i^2 \sigma_i^2 + \sum_{i,j=1, i \neq j}^{N} w_i w_j \mathrm{Cov}(i,j) \right), \quad \sum_{i=1}^{N} w_i = 1$$

To examine how a portfolio with many risky assets works and the ways in which we can reduce the risk of a portfolio, assume that the portfolio has equal weights ($1/N$) for all N assets. In addition, assume that $\overline{\sigma^2}$ and $\overline{\mathrm{Cov}}$ are the average variance and average covariance. Given equal weights and average variance/covariance, we can rewrite the portfolio variance as below (intermediate steps are omitted to focus on the main result):

$$\sigma_P^2 = \left(\sum_{i=1}^{N} w_i^2 \sigma_i^2 + \sum_{i,j=1, i \neq j}^{N} w_i w_j \mathrm{Cov}(i,j) \right)$$

$$\sigma_P^2 = \frac{\overline{\sigma^2}}{N} + \frac{(N-1)}{N} \overline{\mathrm{Cov}}$$

The equation in the second line shows that as N becomes large, the first term on the right side with the denominator of N becomes smaller and smaller, implying that the contribution of one asset's variance to portfolio variance gradually becomes negligible. The second term, however, approaches the average covariance as N increases. It is reasonable to say that for portfolios with a large number of assets, covariance among the assets accounts for almost all of the portfolio's risk.

Importance of Correlation in a Portfolio of Many Assets

The analysis becomes more instructive and interesting if we assume that all assets in the portfolio have the same variance and the same correlation among assets. In that case, the portfolio risk can then be rewritten as:

$$\sigma_p = \sqrt{\frac{\overline{\sigma^2}}{N} + \frac{(N-1)}{N} \rho \sigma^2}$$

The first term under the root sign becomes negligible as the number of assets in the portfolio increases leaving the second term (correlation) as the main determining factor for portfolio risk. If the assets are unrelated to one another, the portfolio can have close to zero risk. In the next section, we review these concepts to learn how portfolios can be diversified.

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A question, if the correlation is negative for large enough number of assets, how do we calculate the square root ?

If the second term converges to a negative number for large enough sample of assets in that case we will end up with a negative number under

NV done

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AZ error! questions from part 2 are included here!

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